SM015 MATHEMATICS 1
SESSION 2024/2025
ANSWER SCHEME – PKA: PECUTAN PSPM SET 1

No.	Section A
1	(a) $\lim_{x\to 0}\left[\frac{2-\sqrt{x+4}}{x(x+1)}\right]=\lim_{x\to 0}\left[\frac{2-\sqrt{x+4}}{x(x+1)}\right]\left[\frac{2+\sqrt{x+4}}{2+\sqrt{x+4}}\right]$ $=\lim_{x\to 0}\left[\frac{4-(x+4)}{x(x+1)\left(2+\sqrt{x+4}\right)}\right]$ $=\lim_{x\to 0}\left[\frac{-x}{x(x+1)\left(2+\sqrt{x+4}\right)}\right]$ $=\lim_{x\to 0}\left[\frac{-1}{(x+1)\left(2+\sqrt{x+4}\right)}\right]$ $=\frac{-1}{(0+1)\left(2+\sqrt{0+4}\right)}=-\frac{1}{4}$
	(b) $\lim_{x\to 2^+}\frac{\lvert 4-2x\rvert}{8-2x-x^2}$ $\quad\lvert 4-2x\rvert=\begin{cases}4-2x, & x\le 2\\ -(4-2x), & x>2\end{cases}$ $=\lim_{x\to 2^+}\frac{-(4-2x)}{8-2x-x^2}$ $=\lim_{x\to 2^+}\frac{-4+2x}{(4+x)(2-x)}$ $=\lim_{x\to 2^+}\frac{-2(2-x)}{(4+x)(2-x)}$ $=\lim_{x\to 2^+}\frac{-2}{(4+x)}$ $=\frac{-2}{4+2}=-\frac{1}{3}$
2	(a) $f(x)=\frac{1}{1-x}$ $f(x+h)=\frac{1}{1-(x+h)}$ $f'(x)=\lim_{h\to 0}\left(\frac{1}{1-(x+h)}-\frac{1}{1-x}\right)\left(\frac{1}{h}\right)$ $=\lim_{h\to 0}\left(\frac{(1-x)-(1-[x+h])}{(1-[x+h])(1-x)}\right)\left(\frac{1}{h}\right)$

$$=\lim_{h\to 0}\left(\frac{1-x-1+x+h}{\left(1-[x+h]\right)(1-x)}\right)\left(\frac{1}{h}\right)$$

$$=\lim_{h\to 0}\left(\frac{h}{\left(1-[x+h]\right)(1-x)}\right)\left(\frac{1}{h}\right)=\lim_{h\to 0}\left(\frac{1}{\left(1-[x+h]\right)(1-x)}\right)$$

$$=\frac{1}{\left(1-[x+0]\right)(1-x)}=\frac{1}{(1-x)^2}$$

(b) $y=\dfrac{\sin x}{1+\cos x}$

$$\frac{dy}{dx}=\frac{(1+\cos x)\cos x-\sin x(-\sin x)}{(1+\cos x)^2}$$

$$=\frac{\cos x+\cos^2 x+\sin^2 x}{(1+\cos x)^2}$$

$$=\frac{\cos x+1}{(1+\cos x)^2}=\frac{1}{1+\cos x}$$

When $x=0:\dfrac{dy}{dx}=\dfrac{1}{1+\cos 0}=\dfrac{1}{2}$

3

$f(x)=x^4-4x^3$

$f'(x)=4x^3-12x^2$

$f'(x)=0$

$4x^3-12x^2=0$

$4x^2(x-3)=0$

$x=0, x=3$

critical numbers are $x=0,\ x=3$

When $x=0; y=0$

when $x=3; y=-27$

The stationary points are $(0,0)$ and $(3,-27)$

$f''(x)=12x^2-24x$ $\qquad$ $f''(x)=12x^2-24x$

$f''(0)=0$ $\qquad$ $f''(3)=12(3)^2-24(3)=36>0$

Interval	$x<0$	$x>0$
Test value	-1	1
Sign of $f'(x)$	-	-
Conclusion	\	\

The stationary point $(0,0)$ is a point of inflection.

$f''(3)=36>0$

The stationary point $(3,-27)$ is a minimum point.

No.	Section
1	(a) $zw = 14 - 2i$ $w = \frac{14-2i}{-3+4i}\left(\frac{-3-4i}{-3-4i}\right)$ $= \frac{-42-56i+6i+8i^2}{9-16i^2}$ $= \frac{-50-50i}{25}$ $= -2-2i$ (b) $\arg(w) = -\left(\pi - \tan^{-1}\left(\frac{2}{2}\right)\right)$ $= -\left(\pi - \frac{\pi}{4}\right)$ $= -\frac{3}{4}\pi$
2	(a) $2^{2x} - 9(2^{x-2}) + \frac{1}{2} = 0$ $(2^x)^2 - \frac{9(2^x)}{2^2} + \frac{1}{2} = 0$ $(2^x)^2 - \frac{9(2^x)}{4} + \frac{1}{2} = 0$ $4(2^x)^2 - 9(2^x) + 2 = 0$ Let $p = 2^x$ $4p^2 - 9p + 2 = 0$ $(4p-1)(p-2) = 0$ $p = \frac{1}{4}$ or $p = 2$ $2^x = \frac{1}{4}$ $\quad$ $2^x = 2$ $2^x = 2^{-2}$ $\quad$ $2^x = 2^1$ $x = -2$ $\quad$ $x = 1$
	(b) $\frac{-15-5x}{x-2} \geq x+3$ $\frac{-15-5x}{x-2} - (x+3) \geq 0$ $\frac{-15-5x-(x+3)(x-2)}{x-2} \geq 0$ $\frac{x^2+6x+9}{x-2} \leq 0$

	$$\frac{(x+3)(x+3)}{x-2} \le 0$$ $$\frac{(x+3)^2}{x-2} \le 0$$ Since $(x+3)^2$ is always positive, then $x-2<0$. $\therefore \{x : x < 2\}$
3	(a) (i) DIY (ii) $R_f = [-1, \infty)$ (iii) $(f \circ f)(-1) = f(-4(-1)-1) = f(3)$ $f(3) = e^{2(3)} - 1$ $= e^6 - 1$
	(b) $(f \circ g)(x) = f(4-3x)$ $= (4-3x)^2 + a(4-3x) + b$ $(f \circ g)(2) = -5$ $(4-3(2))^2 + a(4-3(2)) + b = -5$ $-2a + b = -9 \quad \to (1)$ $(g \circ f)(x) = g(x^2 + ax + b)$ $= 4 - 3(x^2 + ax + b)$ $(g \circ f)(2) = -29$ $4 - 3(x^2 + ax + b) = -29$ $2a + b = 7 \quad \to (2)$ Solve (1) and (2): $a = 4,\ b = -1$
4	(a) $R_f : \left[\frac{1}{2}, \infty\right)$
	(b) $f \circ f^{-1}(x) = x$ $\frac{1}{2}\sqrt{f^{-1}(x) - 4} = x$ $\sqrt{f^{-1}(x) - 4} = 2x$ $f^{-1}(x) - 4 = (2x)^2$ $f^{-1}(x) = 4x^2 + 4$ $D_{f^{-1}} : \left[\frac{1}{2}, \infty\right)$ $R_{f^{-1}} : [5, \infty)$
	(c) DIY

5	(a) $\lim_{x\to\infty}\left(\frac{3x^2+7x-1}{x^2+5}\right)^{\frac{1}{2}}=\lim_{x\to\infty}\left(\frac{\frac{3x^2}{x^2}+\frac{7x}{x^2}-\frac{1}{x^2}}{\frac{x^2}{x^2}+\frac{5}{x^2}}\right)^{\frac{1}{2}}$ $=\lim_{x\to\infty}\left(\frac{3+\frac{7}{x}-\frac{1}{x^2}}{1+\frac{5}{x^2}}\right)^{\frac{1}{2}}$ $=\left(\frac{3+0-0}{1+0}\right)^{\frac{1}{2}}$ $=\sqrt{3}$
	(b) (i) $\lim_{x\to0^+}f(x)=\lim_{x\to0^+}\left(\frac{x}{\sqrt{1+x}-1}\right)\left(\frac{\sqrt{x+1}+1}{\sqrt{x+1}+1}\right)$ $=\lim_{x\to0^+}\frac{x\left(\sqrt{1+x}+1\right)}{(1+x)-1}$ $=\lim_{x\to0^+}\sqrt{1+x}+1$ $=\sqrt{1+0}+1=2$ $\lim_{x\to0^-}f(x)=\lim_{x\to0^-}\left(\frac{4}{\sqrt{4-x}}\right)\left(\frac{\sqrt{4-x}}{\sqrt{4-x}}\right)$ $=\lim_{x\to0^-}\frac{4\sqrt{4-x}}{4-x}$ $=\frac{4\sqrt{4-0}}{4-0}=\frac{4\sqrt{4}}{4}=2$ $\therefore\lim_{x\to0^-}f(x)=\lim_{x\to0^+}f(x)=2$ $\therefore\lim_{x\to0}f(x)$ exist. (ii) (a) $f(0)=\sqrt{2}$ defined (b) $\lim_{x\to0}f(x)$ exist (c) $f(0)\neq\lim_{x\to0}f(x)$ $\therefore$ $f(x)$ is not continous at $x=0$.
6(a)	(a)(i) $e^y=\frac{x^2+3}{x-1},\ x>1$ $\frac{d}{dx}\left(e^y\right)=\frac{d}{dx}\left(\frac{x^2+3}{x-1}\right)$ $e^y\frac{dy}{dx}=\frac{(x-1)(2x)-\left(x^2+3\right)}{(x-1)^2}$

	$e^y \dfrac{dy}{dx} = \dfrac{2x^2-2x-x^2-3}{(x-1)^2}$ $\dfrac{dy}{dx} = \left(\dfrac{1}{e^y}\right)\dfrac{(x^2-2x-3)}{(x-1)^2}$ $\dfrac{dy}{dx} = \left(\dfrac{x-1}{x^2+3}\right)\dfrac{(x-3)(x+1)}{(x-1)^2}$ $\dfrac{dy}{dx} = \dfrac{(x-3)(x+1)}{(x^2+3)(x-1)}$ $\therefore$ shown (ii) When $\dfrac{dy}{dx}=0$: $\dfrac{(x-3)(x+1)}{(x^2+3)(x-1)}=0$ $x=3, x=-1 \quad x>1$ When $x=3$: $e^y = \dfrac{3^2+3}{3-1}$ $e^y=6$ $y=\ln 6$ $\therefore (3,\ln 6)$
6	(b) (i) $x=\sec\theta=(\cos x)^{-1}$ $y=\ln(1+\cos 2\theta)$ $\dfrac{dx}{d\theta}=\dfrac{\sin\theta}{\cos^2\theta}$ $\dfrac{dy}{d\theta}=\dfrac{-2\sin 2\theta}{1+\cos 2\theta}$ $\dfrac{dy}{dx}=\left(\dfrac{-2\sin 2\theta}{1+\cos 2\theta}\right)\left(\dfrac{\cos^2\theta}{\sin\theta}\right)$ $=\left(\dfrac{-2(2\sin\theta\cos\theta)}{1+(2\cos^2\theta-1)}\right)\left(\dfrac{\cos^2\theta}{\sin\theta}\right)$ $=\left(\dfrac{-4(\sin\theta\cos\theta)}{2\cos^2\theta}\right)\left(\dfrac{\cos^2\theta}{\sin\theta}\right)$ $=-2\cos\theta$ $\therefore$ shown (ii) $\dfrac{d^2y}{dx^2}=\dfrac{d}{d\theta}(-2\cos\theta)\times\left(\dfrac{\cos^2\theta}{\sin\theta}\right)$

	$=\left(2\sin\theta\right)\times\left(\dfrac{\cos^2\theta}{\sin\theta}\right)=2\cos^2\theta$ When $\theta=\dfrac{\pi}{3}$: $\dfrac{d^2y}{dx^2}=2\left(\cos\dfrac{\pi}{3}\right)^2$ $=2\left(\dfrac{1}{2}\right)^2=\dfrac{1}{2}$
7	(a) $V=\pi r^2h$ $\pi r^2h=54\pi$ $h=\dfrac{54}{r^2}$ $S=2\pi rh+2\pi r^2$ $S=2\pi r\left(\dfrac{54}{r^2}\right)+2\pi r^2$ $S=\dfrac{108\pi}{r}+2\pi r^2$ $\therefore$ shown
	(b) Minimum/maksimum surface area occurs at $\dfrac{ds}{dr}=0$. $-\dfrac{108\pi}{r^2}+4\pi r=0$ $4\pi r^3=108\pi$ $r^3=27$ $r=3$ Second derivative test: $\dfrac{d^2s}{dr^2}=\dfrac{216\pi}{r^3}+4\pi$ When : $\dfrac{d^2s}{dr^2}=\dfrac{216\pi}{3^3}+4\pi$ $=12\pi>0$ So, surface area is minimum when $r=3cm$